# Hyperbolas

## Standard form

1 For the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, find:

**a** the vertices

**b** the foci

**c** the equations of the asymptotes

2 Write the standard-form equation of the hyperbola with vertices $(0, \pm 2)$ and foci $(0, \pm 3)$.

3 Consider the equation $4x^2 - y^2 + 8x + 4y - 4 = 0$.

**a** Write the equation in standard form by completing the square.

**b** Identify the conic and state its center.

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## Identifying conics

4 Identify each conic section from its equation:

**a** $x^2 + y^2 = 49$

**b** $\frac{x^2}{9} + \frac{y^2}{25} = 1$

**c** $y^2 - 4x^2 = 16$

**d** $y = 3x^2 - 6x + 1$

5 A hyperbola has asymptotes $y = \pm 2x$ and vertices $(\pm 3, 0)$. Find its equation.

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## Hyperbolas centered away from the origin

6 For the hyperbola $\frac{(y+1)^2}{4} - \frac{(x-3)^2}{9} = 1$, find:

**a** the center

**b** the vertices

**c** the direction of opening

7 Complete the square to write $9x^2 - 4y^2 - 18x - 16y - 43 = 0$ in standard form, and state the center.

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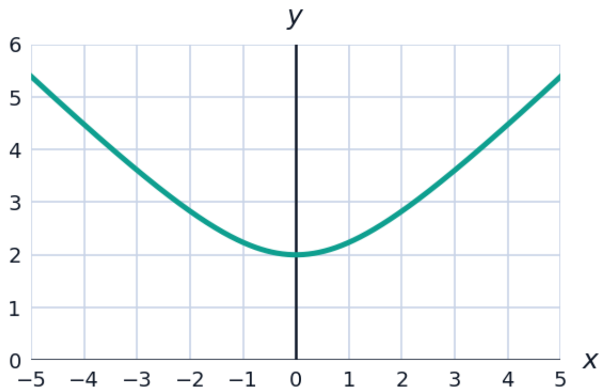
## Eccentricity and applications

8 Find the eccentricity of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$, to two decimal places, and state whether it is closer to a pair of straight lines or to a circle.

9 A ship uses two radio towers 200 km apart (foci) to navigate via hyperbolic positioning: the difference in distances from the ship to each tower is constant at 120 km. Write the equation of the hyperbola (centered at the midpoint of the towers, opening left-right), in standard form.

## Graphing a hyperbola branch

10 The upper-right branch of $y^2 - x^2 = 4$ (as $y = \sqrt{4 + x^2}$) is shown. Use the graph to estimate $y$ at $x = 3$, then verify algebraically, and compare to the asymptote $y = x$.




## The hyperbola's defining property

11 State the geometric (difference-of-distances) definition of a hyperbola, and use it to verify the vertex $(4, 0)$ of $\frac{x^2}{16} - \frac{y^2}{9} = 1$ (foci $(\pm 5, 0)$) satisfies it.

## Comparing all four conic sections

12 State the key algebraic feature (coefficient signs) that distinguishes a circle, ellipse, parabola, and hyperbola in a general second-degree equation $Ax^2 + By^2 + \ldots = 0$.

## Finding the equation from asymptotes and a point

13 A hyperbola centered at the origin has asymptotes $y = \pm\frac{2}{3}x$ and passes through $(3, 0)$. Find its equation.

## Applications: LORAN navigation

**14** Two LORAN radio towers are 260 km apart. A ship determines the difference in distances to the two towers is 100 km. Write the equation of the hyperbola of possible ship locations (centered at the midpoint, opening along the line connecting the towers).